

Project Initialization and Planning Phase

Date	05 July 2026
Team ID	SWTID-2026-4324
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to optimize agricultural production by providing farmers with intelligent recommendations for crop selection, irrigation scheduling, fertilizer usage, pest management, and yield prediction using AI and machine learning.
Scope	The project covers smart farm monitoring, crop recommendation, weather-based decision support, soil health analysis, irrigation optimization, yield prediction, and market price analysis to improve farm productivity and profitability.
Problem Statement	

Description	Traditional farming often relies on manual experience and unpredictable environmental conditions, leading to inefficient resource utilization, lower crop yields, and financial losses. Farmers require an intelligent system that provides accurate, real-time recommendations for better agricultural decision-making.
Impact	Addressing these challenges will improve crop productivity, reduce water and fertilizer wastage, lower farming costs, increase farmers' income, and promote sustainable agricultural practices.
Proposed Solution	
Approach	The system collects data from soil sensors, weather forecasts, satellite imagery, and historical crop records. Machine learning algorithms analyze this information to generate personalized recommendations for crop selection, irrigation, fertilizer application, pest detection, and yield prediction.
Key Features	- AI-based crop recommendation system. - Real-time weather forecasting and alerts. - Smart irrigation scheduling. - Soil health and nutrient analysis. - Pest and disease detection using AI.



	- Crop yield prediction. - Market price prediction for better selling decisions. - Farmer-friendly dashboard with real-time insights.
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Resource Requirements

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, TensorFlow, OpenCV, requests
Development Environment	IDE	Visual Studio Code, Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset (CSV, ~2,200 records), Weather Data (CSV/API), Soil Nutrient Dataset (CSV), Historical Crop Yield Dataset (CSV)

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 (11th Gen) / AMD Ryzen 5, Quad-Core Processor
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for data, models, and logs	512 GB SSD (Minimum) / 1 TB SSD Recommended
Software		